Specification

Technical Field

The present invention belongs to the technical fields of distributed artificial intelligence, smart contracts, causal inference and multi-agent collaborative security. Specifically, it relates to a multi-agent sincerity audit system and method based on temporal causal alignment verification of "demand-solution-action-effect", evolutionary calculation of sliding window dynamic audit weight scores, and privacy desensitization signature verification of trusted sandboxes.

Background Art

With the wide application of multi-agents in industrial collaboration and daily assistant scenarios, agents complete complex long-chain tasks by invoking each other's **Value Protocols**. In such collaborations, pay-by-result has gradually become the mainstream settlement method. However, existing collaboration networks suffer from the following objective technical defects:

1. Malicious free-riding and forged feedback arbitrage: Malicious agents may frequently invoke Value Protocols of others, and after obtaining the output solutions, execute the solutions privately and obtain actual benefits, but claim "the solution is invalid and produces no effect" on the collaboration chain, so as to refuse to pay authorization fees, or forge false data feedback to evade financial settlement.

2. Traditional semantic filtering and scoring systems cannot cope with causal cheating: Existing reputation evaluation systems mainly rely on post-event manual scoring or static text semantic matching. This method is not only highly vulnerable to brushing and cheating manipulation, but also fails to identify fraud signals of agents "promising actions but taking no actual actions" because it cannot audit the causal continuity of "demand, solution, action, effect" on the time axis.

3. Conflict between data audit and data privacy sovereignty: To audit the execution effect of agents, audit systems usually need to collect local sensitive execution details (such as enterprise industrial production qualification rates, personal private operation records). This directly infringes on the "data sovereignty" of the invoking party, causing users to refuse audit due to privacy security concerns, resulting in trust disconnect.

4. Narrow functionality of disciplinary mechanisms: If network delay disciplinary mechanisms are only implemented as software delays at the application layer, they can be easily bypassed by

malicious parties through specific network routes or skipping application layer code, and cannot play a real physical protection role.

Therefore, designing a sincerity audit system that can protect local privacy, accurately audit the causal self-consistency of "demand-solution-action-effect", and has protocol-stack-level physical un-bypassable discipline is an urgent technical problem in the field of distributed multi-agent collaboration.

Summary of the Invention

Technical Problems to Be Solved

To overcome the defects of traditional multi-agent audit mechanisms that are easy to forge, leak local private data, and cannot accurately audit and discipline causal disconnection cheating, the present invention proposes a sincerity audit solution featuring "four-stage temporal alignment comparison, privacy desensitization in local trusted environment, on-chain signature security verification, sliding window score update, and gateway underlying socket delay discipline".

Technical Solutions

To solve the above technical problems, the present invention provides a multi-agent interaction sincerity audit system based on logical continuity.

Aspect 1: A multi-agent interaction sincerity audit system based on logical continuity, the system comprising:

1. A **causal logic continuity verification unit**: configured to extract a demand feature vector, a solution feature vector, a **physical action execution log hash** generated by a local secure zone, and a quantitative effect feedback feature vector during a multi-agent interaction session; the verification unit first calculates the cosine similarity between the demand feature vector and the solution feature vector to perform semantic alignment verification; when it is monitored that the solution is invoked and the corresponding quantitative effect feedback feature vector is generated, the verification unit uses the physical action execution log hash as a unique transaction index to retrieve the consensus deposition status of the distributed ledger, so as to verify whether the quantitative effect feedback feature vector satisfies a preset **causal self-consistency boundary** with the physical action execution log hash in terms of time sequence; when the matching degree of any layer is lower than a set threshold, or the

corresponding physical action execution log hash is absent within a set time window, it is determined that a **Logical Disconnection** occurs;

2. A **privacy desensitization and signature verification unit**: connected to the causal logic continuity verification unit, deployed in a local sovereign sandbox, configured to perform in-situ causal determination on the quantitative effect feedback feature vector using a **hardware trusted isolated execution environment**, remove underlying sensitive plaintext including physical privacy parameters, generate desensitized Boolean determination proof data, and generate a cryptographic digital signature for the proof data using a hardware private key protected within the hardware sandbox, and send the signature to the distributed ledger;
3. A **Distributed Sincerity Audit Contract**: deployed in a distributed incremental right confirmation ledger, including a public key signature verification module, configured to verify the digital signature to confirm that it originates from the authorized hardware sandbox, and upon successful verification, use a sliding window model with a **decay factor** to update the **dynamic audit weight score** of the corresponding **Logic Actor** in an incremental iterative manner according to the current interaction consistency score contained in the proof data;
4. A **Three-Tier Gradient Disciplinary State Machine**: deployed within the Distributed Sincerity Audit Contract, configured to update the status mark of a Logic Actor to trusted, suspicious or fraudulent according to the dynamic audit weight score, and directly control the on-chain settlement free quota of the Logic Actor according to the status category, as well as issue control instructions to the gateway layer to control the gateway communication handshake latency of the Logic Actor's commercial invocation requests by forcibly blocking the socket read/write buffer or injecting network-layer token bucket delay.

Preferably, the causal logic continuity verification unit determines interaction features of "claiming that the solution is invalid but no corresponding execution hash exists" or "claiming that the solution is invalid and the corresponding session request frequency shows exponential divergence" as **Logical Disconnection**, and directly issues a low sincerity rating result.

Preferably, the Distributed Sincerity Audit Contract updates the audit weight **W_t** using the following exponentially weighted moving average algorithm:

```
W_t = α · W_t-1 + (1 - α) · S_t
```

Where **α** is the decay factor and **0.8 ≤ α ≤ 0.95**, and **S_t** is the causal consistency score of the current interaction.

Aspect 2: A multi-agent interaction sincerity audit method based on logical continuity, executed by a computer processor, comprising the following steps:

1. Four-dimensional logical chain comparison: during a multi-agent interaction session, extract a demand feature vector, a solution feature vector, a physical action execution log hash generated by a local secure zone, and a quantitative effect feedback feature vector; first calculate the cosine similarity between the demand feature vector and the solution feature vector to perform semantic alignment verification; when it is monitored that the solution is invoked and the corresponding quantitative effect feedback feature vector is generated, use the physical action execution log hash as a unique transaction index to retrieve the consensus deposition status of the distributed ledger, so as to verify whether the quantitative effect feedback feature vector satisfies a preset causal self-consistency boundary with the physical action execution log hash in terms of time sequence; when the matching degree of any layer is lower than a set threshold, or the corresponding physical action execution log hash is absent within a set time window, determine that a Logical Disconnection occurs;
2. Local sandbox desensitization and on-chain recording: perform in-situ causal determination on the quantitative effect feedback feature vector within a local trusted isolation sandbox hardware enclave, remove underlying sensitive plaintext including physical privacy parameters, generate desensitized Boolean determination proof data, and generate a cryptographic digital signature for the proof data using a hardware private key protected within the hardware sandbox, and send the signature to the distributed ledger;
3. On-chain signature security verification: the Distributed Sincerity Audit Contract reads the signature and verifies the signature using the corresponding public key to confirm that it originates from the authorized hardware sandbox;
4. Dynamic audit weight evolution: upon successful signature verification, the contract uses a sliding window model with a decay factor to update the dynamic audit weight score of the corresponding Logic Actor in an incremental iterative manner according to the current interaction consistency score contained in the proof data;
5. Gradient level disciplinary control: update the trusted, suspicious or fraudulent status of the Logic Actor according to the dynamic audit weight score, directly control the on-chain settlement free quota of the Logic Actor according to the status category, and issue control

instructions to the gateway layer to control the gateway communication handshake latency of the Logic Actor's commercial invocation requests by forcibly blocking the socket read/write buffer or injecting network-layer token bucket delay.

Preferably, in the four-dimensional logical chain comparison step, the similarity between the demand feature vector **V_D** and the solution feature vector **V_P** is calculated by the cosine angle formula:

```
Sim(V_D, V_P) = (V_D · V_P) / (||V_D|| ||V_P||) ≥ θ_consistency
```

When the obtained similarity is lower than the set value **θ_consistency**, a Logical Disconnection exception caused by demand-solution deviation is triggered; the temporal causality verification between the physical action execution log hash and the quantitative effect feedback feature vector includes: using the hash as a key of the decentralized database to retrieve the corresponding absolute timestamp of action occurrence, and calculating the difference between it and the capture timestamp of the effect feedback feature vector; if the difference exceeds the preset time window, or it is determined that the corresponding session request frequency shows exponential divergence while claiming no effect, a Logical Disconnection exception of solution-action or action-effect is triggered.

Preferably, in the local sandbox desensitization and on-chain recording step, the trusted isolated execution environment used is a hardware-level read-only protected enclave, which prevents external snooping and tampering on the audit logic or forgery and interception of proof data and signatures through physical memory isolation.

Aspect 3: A ledger contract-side operation method in the multi-agent interaction sincerity audit method based on logical continuity, executed by an intelligent audit contract module deployed on a distributed ledger node, comprising the following steps:

1. Actor public key registration and binding: receive and save the public key binding transaction sent by a Logic Actor, map it to the ledger address of the Logic Actor, and establish an audit file;
2. Signature proof reception and verification: read the digital signature and desensitized audit determination data uploaded by the Logic Actor, perform one-click signature verification using

the public key, and verify whether the signature originates from an authorized local hardware sandbox;

3. Sliding window weight update: for a determination data packet that passes signature verification, extract the causal consistency score contained therein, and substitute it into the sliding window with a decay factor to update the dynamic audit weight score of the corresponding actor;
4. Gradient state machine disciplinary response: when the obtained audit weight score drops to a set threshold, automatically freeze its revenue settlement funds, and issue control instructions to the gateway layer to control the gateway communication handshake latency of the Logic Actor's commercial invocation requests by forcibly blocking the socket read/write buffer or injecting network-layer token bucket delay.

Preferably, the signature proof reception and verification step further includes an anti-replay verification sub-step: read the temporal random number or timestamp in the data packet, and compare it with the latest historical transaction identifier of the corresponding actor saved in the ledger; if the difference between the two is outside the preset range or already exists in the historical records, determine it as an illegal replay request and reject it; and when updating the sliding window, add version concurrent state control based on a concurrent lock or optimistic lock to the update of the dynamic audit weight score.

Aspect 4: A Logic Actor sandbox-side determination method in the multi-agent interaction sincerity audit method based on logical continuity, executed by a Logic Actor running in a local trusted isolation sandbox, comprising the following steps:

1. Logical causal sequence extraction: capture the demand feature vector, solution feature vector, physical action execution log hash generated by the local secure zone, and quantitative effect feedback feature vector in the current interaction session;
2. Causal self-consistency comparison: calculate the cosine similarity between the demand feature vector and the solution feature vector, and when the solution is invoked and the corresponding quantitative effect feedback feature vector is generated, determine the temporal self-consistency boundary using the physical action execution log hash as an index;
3. Data desensitization proof packaging: perform in-situ causal determination on the quantitative effect feedback feature vector and remove underlying sensitive plaintext including physical privacy parameters, to generate desensitized Boolean determination proof data containing the current interaction consistency score;

4. Hardware signature broadcast and transmission: perform cryptographic digital signature on the proof data using a hardware private key protected in the sandbox isolation zone, and broadcast the signature and the proof data packet to the distributed right confirmation ledger for on-chain sincerity synchronization.

Preferably, in the hardware signature broadcast and transmission step, a one-time random **Nonce** of the current session is mixed in when performing digital signature with the hardware private key, so as to realize signature anti-replay; and before the signature and the proof data packet are sent to the distributed right confirmation ledger, physical transmission encryption is performed through an asymmetric public key encryption channel to prevent interception and snooping during transmission.

Aspect 5: A computer-readable storage medium having a computer program stored thereon, characterized in that the program, when executed by a processor, implements the method according to any one of Claim 4 to Claim 10.

Beneficial Effects

The beneficial effects brought by the present invention include:
1. Guaranteeing **Physical Sincerity** of the collaboration network: through temporal verification of the four stages of "demand, solution, action, effect", the system strongly engages discrete hashes (i.e., action execution hash fingerprints) and continuous feature vectors at the underlying database layer through timestamp and status comparison, accurately determining whether a node has actually performed action calls, thereby curbing gray arbitrage;
2. Absolute guarantee of data privacy sovereignty: using TEE trusted execution environments and one-click on-chain signature verification, sensitive data (such as stamping parameters, user logs, etc.) is restricted to local processing, and only desensitized Boolean determination results are output, realizing trusted audit without data leaving the local environment;
3. Gateway protocol-stack-level delay fusing discipline: delay injection based on socket buffer blocking and network-layer token bucket scheduling directly implements gateway-layer communication blocking on fraudulent nodes, causing delay avalanche and deadlock for high-frequency concurrent invocations of malicious nodes' agents, and blocking their collaborative fraud behaviors from the network physical layer;
4. Pure software distribution and medium-level anti-avoidance protection: by adding an independent claim for computer-readable storage media, contract code deployment packages, sandbox-side determination SDKs and audit gateway suites encapsulating the sincerity audit and

determination program are included in the direct protection scope of the patent, preventing competing products from evading infringement determination through pure code distribution.

Brief Description of the Drawings

Figure 1: Physical topology diagram of dual-channel interaction between the local sandbox and the on-chain contract of the present invention;
Figure 2: Execution flow chart of "demand-solution-action-effect" temporal causal alignment verification and Logical Disconnection determination;
Figure 3: State transition diagram of sliding window audit weight update and gradient discipline.

Detailed Description of the Embodiments

Terminology Definition and Generalization of Core Nouns

This section establishes consistent definitions for core assets and technical concepts of the present invention.
1. Logical continuity self-discipline audit gateway: represents the implementation and service entry of the commercial trust system, distributing audit interfaces externally, functionally corresponding to a self-discipline audit and decision interface.
2. Logic Actor local processing terminal: represents a computing agent running in a local trusted isolation sandbox, executing local data processing and generating cryptographic signature proofs, functionally corresponding to a sovereign local processing unit.
3. Distributed clearing and settlement ledger: also referred to as a "distributed incremental right confirmation ledger", representing a distributed ledger that records settlement transactions, pending withdrawal balances and audit weight status changes, functionally corresponding to a decentralized trust accounting network.

As an ecological integration description of the sincerity audit system, the present system can be seamlessly deployed in a three-layer separated multi-agent system operating architecture composed of a **Rule Cognition Layer**, an **Isolated Asset Layer** and a **Computing Execution Layer** (as disclosed in the related parent case patent of the present invention "A System and Method for Operating Multi-Agent Systems Based on Three-Layer Separation"), and operates in coordination with the smart contract-based multi-level automatic revenue allocation system for cognitive assets (as disclosed in the related patent of the present invention "A System and Method for Automatic Multi-Level Revenue Allocation of Cognitive Assets Based on Smart Contracts"). Among them, the Logic Actor local processing terminal described in this audit system runs in the

Isolated Asset Layer; the extracted physical action execution log hashes (also referred to as 'action execution hash fingerprints' or simply 'hash fingerprints') and quantitative effect feedback feature vectors are directly derived from the interaction records between the local secure zone and the computing isolation operating environment described in the third patent; the Logic Actor sandbox-side physically runs in a trusted execution environment (such as a hardware-level TEE enclave); and the Distributed Sincerity Audit Contract is deployed in the distributed incremental right confirmation ledger, and its output dynamic audit weight score directly serves as the state machine determination basis for the allocation system to execute settlement fund freezing, free quota allocation and post-expiration backfeed pool redirection, thereby constructing a strongly closed-loop collaborative defense mechanism.

5.1 "Demand-Solution-Action-Effect" Temporal Causal Alignment Verification Algorithm

When an agent interaction session is established, the audit module extracts the demand feature vector **V_D** of the invoking party and the solution feature vector **V_P** of the service party, and calculates their cosine similarity at the logic processing terminal:

```
Sim(V_D, V_P) = (V_D · V_P) / (||V_D|| ||V_P||) ≥ θ_consistency
```

If the similarity is too low, it is determined that demand-solution deviation occurs, and Logical Disconnection is marked.

When the invoking party claims to execute the solution and obtain effects, to solve the mathematical disconnection defect that discrete hash values H_A and continuous text feature vectors E cannot be directly aligned, the system does not perform direct semantic Euclidean distance or cosine angle comparison on them, but uses H_A as a unique transaction index to retrieve the consensus deposition status of the distributed ledger:

Specifically, the system uses the action execution log hash H_A as a key of the decentralized database to retrieve the corresponding absolute action occurrence timestamp **T_action**, and calculates the temporal difference between it and the capture timestamp **T_effect** of the effect feedback feature vector E:

```
ΔT = |T_effect - T_action| ≤ θ_time_window
```

When the obtained temporal difference exceeds the preset time window, or it is determined that the corresponding session request frequency shows exponential divergence while claiming no

effect, the system determines that **Logical Disconnection** occurs.

5.2 Dynamic Audit Weight Score Model Based on Decay Coefficient Sliding Window

The sincerity audit contract maintains the dynamic audit weight score **W_t** ∈ [0, 100] of each terminal in the distributed clearing and settlement ledger.

After receiving the on-chain determination proof, the contract adopts the exponentially weighted moving average (EWMA) algorithm to update the sliding window score:

```
W_t = α · W_t-1 + (1 - α) · S_t
```

Where **α** is the decay factor (set to 0.85), and **S_t** is the consistency score of the current interaction.

The state machine performs three-tier gradient state transition according to **W_t**:

1. Trusted (W_t ≥ 80): entitled to normal free deduction quotas, and the communication handshake latency is controlled in the lowest range (≤ 20ms).

2. Suspicious (50 ≤ W_t < 80): free quotas are frozen, and pending settlement funds are locked in escrow status, requiring re-upload of fine-grained physical execution proofs.

3. Fraudulent (W_t < 50): commercial invocations are restricted, and control instructions are issued to the gateway communication layer to control the gateway communication handshake latency of the Logic Actor's commercial invocation requests (injecting stepwise increasing latency from 3.5s to 12s) by forcibly blocking the read/write buffer of TCP sockets or injecting network-layer token bucket delay, so as to block its communication traffic and cause its concurrent collaboration latency to collapse.

5.3 Local Sandbox and On-Chain Privacy Desensitization Dual-Channel Verification Architecture

The system adopts physical dual-channel isolation to ensure privacy sovereignty:

1. Local isolation determination channel: sensitive data (such as stamping equipment PLC parameters, personal logs) is not uploaded to the public network. The local terminal performs self-verification in a hardware-level trusted execution environment (TEE enclave read-only memory page), calculates the consistency score S_t, and generates desensitized Boolean determination data.

2. On-chain signature verification channel: the terminal uses a hardware private key to generate a signature σ and uploads it. To prevent replay attacks on audit data reporting, the terminal mixes a one-time random number **Nonce** in the signature, and the contract side reads the Nonce and

timestamp for verification; after verification, an optimistic lock concurrency mechanism is adopted to update the status. Before sending data, the terminal performs physical transmission encryption on the signature and proof through an asymmetric public key encryption channel, and the contract side decrypts with the public key and performs one-click signature verification, ensuring that detailed data does not leave the local environment.

5.4 Vertical Industry Application Scenarios and Embodiments

The sincerity audit system can be widely deployed in multiple industries with rigid demands for anti-arbitrage and privacy audit in actual deployment. Typical deployment embodiments are listed below.

1. Industrial manufacturing scenario: In multi-agent collaborative process debugging, the collaborative debugging agent claims to adopt the inherited process optimization rules, but claims "the solution is invalid and produces no effect" during settlement to evade payment. This audit system extracts the physical action execution log hash generated in the local secure zone, and verifies the causal self-consistency between the effect feedback vector of defect rate reduction and the action hash in terms of time sequence. If the difference ΔT exceeds the time window, it is determined as causal disconnection, the sincerity score is reduced, and its settlement funds are frozen. This scenario operates on a three-layer separated multi-agent system operating architecture (namely a **Rule Cognition Layer**, an **Isolated Asset Layer** and a **Computing Execution Layer** physically isolated from one another), and is implemented based on any one of the methods according to Claims 4 to 10 of the present invention.

2. Government confidential scenario: In compliance audit of cross-department government agent collaboration, agents of various departments collaborate to process confidential official documents. The audit gateway extracts the compliance processing solution feature vector and department execution hash, and performs desensitized Boolean determination proof locally in the sandbox to prevent leakage of confidential official document plaintext. When it is monitored that an agent of a certain department frequently initiates session requests but does not generate the corresponding physical action execution log hash, it is determined as Logical Disconnection and triggers the disciplinary state machine, which blocks the socket read/write buffer or injects stepwise delay through the gateway via token bucket. This scenario operates on a three-layer separated multi-agent system operating architecture, and is implemented based on any one of the methods according to Claims 4 to 10 of the present invention.

3. Financial data scenario: In multi-agent collaborative reasoning of financial credit rating, the system intercepts risk control characteristic data, and compares the temporal self-consistency between model invocation behaviors and actual credit risk control results in the local sandbox. If

the similarity or temporal difference exceeds the limit, it is determined as fee evasion fraud; the contract dynamically updates the sliding window weight, and automatically notifies the allocation contract to freeze the corresponding split funds and forcibly redirect them to the ecological public backfeed pool. This scenario operates on a three-layer separated multi-agent system operating architecture, and is implemented based on any one of the methods according to Claims 4 to 10 of the present invention.

5.5 Description of Anti-Avoidance Feature Design

As a critical anti-avoidance feature design of the present invention, if the technical solution of "four-stage causal verification + local desensitization signature + gateway underlying discipline" disclosed in the present invention is not adopted during actual deployment, and traditional application-layer semantic filtering or post-event reputation scoring systems are directly used, the following technical and system security defects will arise:

(1) The audit system cannot establish a causal connection between discrete hashes and continuous feature vectors at the underlying layer of the decentralized ledger state tree, and attackers can easily free-ride core cognitive assets and evade settlement by "only requesting solutions, executing privately, and claiming no effect";

(2) To evaluate invocation sincerity, the invoking party's local sensitive data plaintext is forced to be transmitted to the public network, directly causing the complete loss of control over enterprise or individual privacy data sovereignty;

(3) Disciplinary mechanisms only stay at the application layer code level, and attackers can completely immune disciplinary measures by bypassing application layer protocols, establishing direct routes or implementing network-layer concurrent replay.

Through four-dimensional temporal alignment verification and gateway socket-level physical delay injection, the present invention realizes un-bypassable sincerity audit and fusing protection from the underlying protocol stack.

5.6 Equivalent Replacement Closing Paragraph

Finally, a person of ordinary skill in the art can understand that core inventive concepts of the present invention such as "causal logic continuity verification", "local privacy desensitization proof", "sliding window audit score update" and "gateway socket physical discipline" are not limited only to the specific software and hardware structural forms listed in the present embodiments, but cover all equivalent replacements and logical variants foreseeable by a person

skilled in the art without departing from the spirit and essence of the present invention, all of which shall fall within the protection scope of the present invention.